

Maintenance Schedule

Diesel Engine

10V1600T70

12V1600T50

12V1600T70

Application Group 4B

MS50219/00E



Power. Passion. Partnership.

Engine model	kW/cyl.	Application group
10V1600T70	61 kW/cyl.	4B, Continuous operation, variable
12V1600T50	53 kW/cyl.	4B, Continuous operation, variable
12V1600T70	61 kW/cyl.	4B, Continuous operation, variable

Table 1: Applicability

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1 Maintenance Schedule

1.1 Preface

In the case of emission-certified products, nonobservance of the maintenance specifications can result in a violation of the statutory emission regulations. Modification or removal of mechanical or electronic components or the installation of additional components as well as the execution of calibration processes that might affect the emission characteristics of the product are prohibited by emission regulations. Emission-relevant components may only be maintained, exchanged or repaired if the components used for this purpose are approved by MTU or equivalent components. Emission-relevant components are for instance control units, data records, sensors, injectors, exhaust flaps and all components of exhaust aftertreatment systems.

The maintenance system for MTU products is based on a preventive maintenance concept. Preventive maintenance facilitates advance planning and ensures a high degree of equipment availability. The maintenance intervals and required maintenance scopes are based on operational experience and must therefore be regarded as recommendations. In the event of difficult operational and ambient conditions, additional maintenance tasks and/or changes to the maintenance intervals may be necessary.

Maintenance activities are specified with intervals based on hours in operation and time limits. The limit which occurs first is applicable.

The individual maintenance intervals are assigned to the qualification levels QL1 to QL4.

QL1: Operational checks and maintenance work that does not require disassembly of the product.

QL2: Component replacement (corrective, and thus not an element of the maintenance schedule).

QL3: Maintenance work that requires partial disassembly of the product.

QL4: Maintenance work that requires complete disassembly of the product.

Supplementary information on maintenance and preservation:

Change intervals for fluids and lubricants and details on preservation can be obtained in the relevant specifications of the component manufacturers and MTU Fluids and Lubricants Specifications. The latest version is also available at: <http://www.mtu-online.com>, for engine-generator sets at: <http://www.mtuonsiteenergy.com>.

Components which are not listed in this maintenance schedule must be serviced in accordance with Manufacturer's Instructions.

This maintenance schedule does not apply to engines with an EPA/CARB emission certificate.

1.2 Allocation matrix application group 4B

Load profile		10V/12V1600Tx0						
Load %	Time %	Load factor %	Load indicator %	0 to 55 kW/cyl.	60 to 65 kW/cyl.			
110 ₍₁₀₀₋₁₁₀₎	—	0 to 31	≤ 17	18,000	15,000	—	—	Standard
100 ₍₉₀₋₁₀₀₎	12							
90 ₍₈₀₋₉₀₎	12							
80 ₍₆₅₋₈₀₎	—							
65 ₍₅₀₋₆₅₎	—							
50 ₍₃₀₋₅₀₎	—							
30 ₍₅₋₃₀₎	16							
Idle Speed ₍₀₋₅₎	60							
110 ₍₁₀₀₋₁₁₀₎	—	32 to 35	≤ 19	18,000	15,000	—	—	
100 ₍₉₀₋₁₀₀₎	17							
90 ₍₈₀₋₉₀₎	—							
80 ₍₆₅₋₈₀₎	13							
65 ₍₅₀₋₆₅₎	—							
50 ₍₃₀₋₅₀₎	—							
30 ₍₅₋₃₀₎	15							
Idle Speed ₍₀₋₅₎	55							
110 ₍₁₀₀₋₁₁₀₎	—	36 to 45	≤ 32	15,000	12,000	—	—	
100 ₍₉₀₋₁₀₀₎	26							
90 ₍₈₀₋₉₀₎	15							
80 ₍₆₅₋₈₀₎	—							
65 ₍₅₀₋₆₅₎	—							
50 ₍₃₀₋₅₀₎	—							
30 ₍₅₋₃₀₎	12							
Idle Speed ₍₀₋₅₎	47							
110 ₍₁₀₀₋₁₁₀₎	—	46 to 55	≤ 41	12,000	9,000	—	—	
100 ₍₉₀₋₁₀₀₎	35							
90 ₍₈₀₋₉₀₎	15							
80 ₍₆₅₋₈₀₎	—							
65 ₍₅₀₋₆₅₎	—							
50 ₍₃₀₋₅₀₎	—							
30 ₍₅₋₃₀₎	15							
Idle Speed ₍₀₋₅₎	35							

Table 2: Allocation matrix application group 4B

TIM-ID: 0000066779 - 001

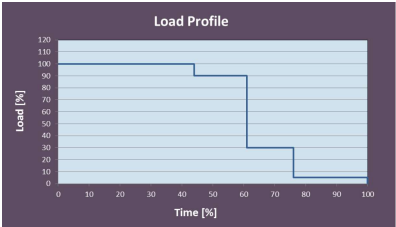
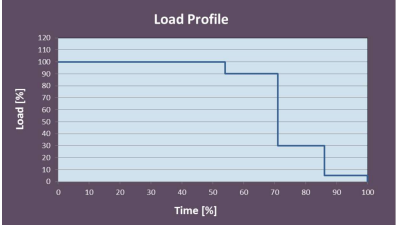
10V/12V1600Tx0								
Load profile		Load factor %	Load indicator %	0 to 55 kW/cyl.	60 to 65 kW/cyl.			
Load %	Time %							
110 ₍₁₀₀₋₁₁₀₎	—	56 to 65	≤ 51	9,000	6,000	—	—	
100 ₍₉₀₋₁₀₀₎	44							
90 ₍₈₀₋₉₀₎	17							
80 ₍₆₅₋₈₀₎	—							
65 ₍₅₀₋₆₅₎	—							
50 ₍₃₀₋₅₀₎	—							
30 ₍₅₋₃₀₎	15							
Idle Speed ₍₀₋₅₎	24							
110 ₍₁₀₀₋₁₁₀₎	—	66 to 75	≤ 61	6,000	—	—	—	
100 ₍₉₀₋₁₀₀₎	54							
90 ₍₈₀₋₉₀₎	17							
80 ₍₆₅₋₈₀₎	—							
65 ₍₅₀₋₆₅₎	—							
50 ₍₃₀₋₅₀₎	—							
30 ₍₅₋₃₀₎	15							
Idle Speed ₍₀₋₅₎	14							

Table 3: Allocation matrix application group 4B